Introduction to multi-omics data integration

Ahmed Mahfouz

Human Genetics, Leiden University Medical Center
Pattern Recognition and Bioinformatics, TU Delft

Leids Universitair Medisch Centrum

mahfouzlab.org

TU Delft

What we cover in this workshop

- What is multi-omics data?
- What is high-dimensional data?
- What are the different ways to analyze high dimensional, multimodal data?
- Brief introduction to (multimodal) single-cell genomics

Age of Big Data

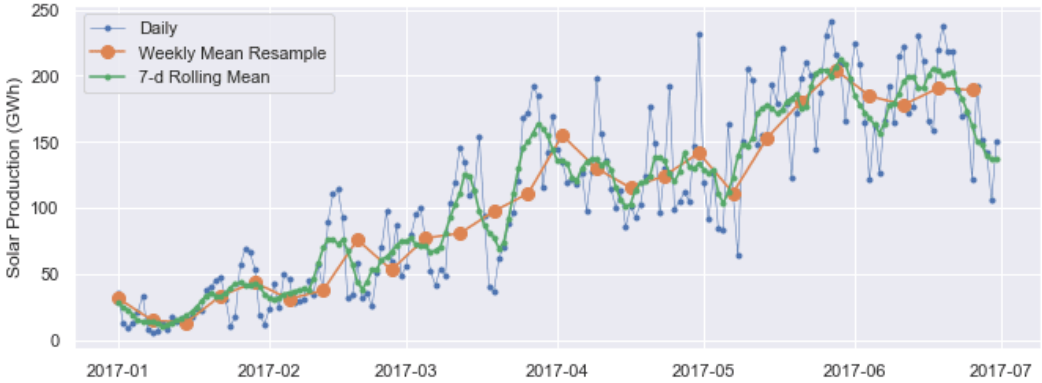
Images

Videos

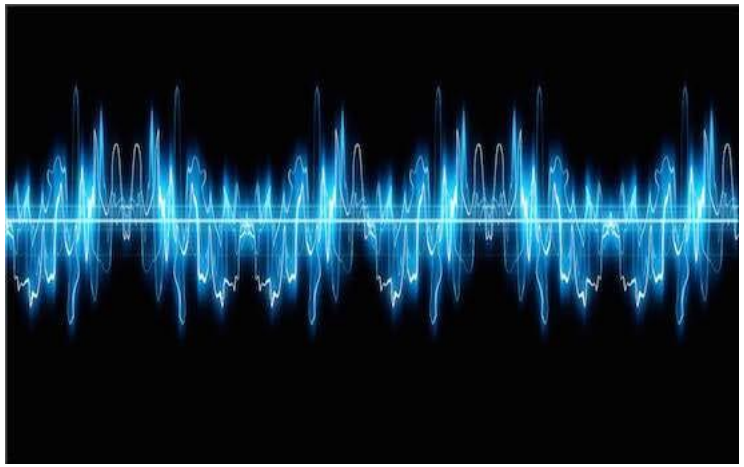
Text

WIKIPEDIA
The Free Encyclopedia

Time series

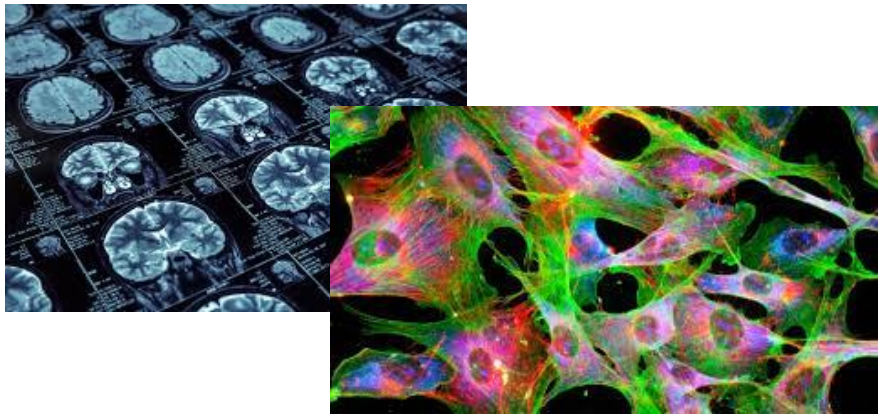


Sound

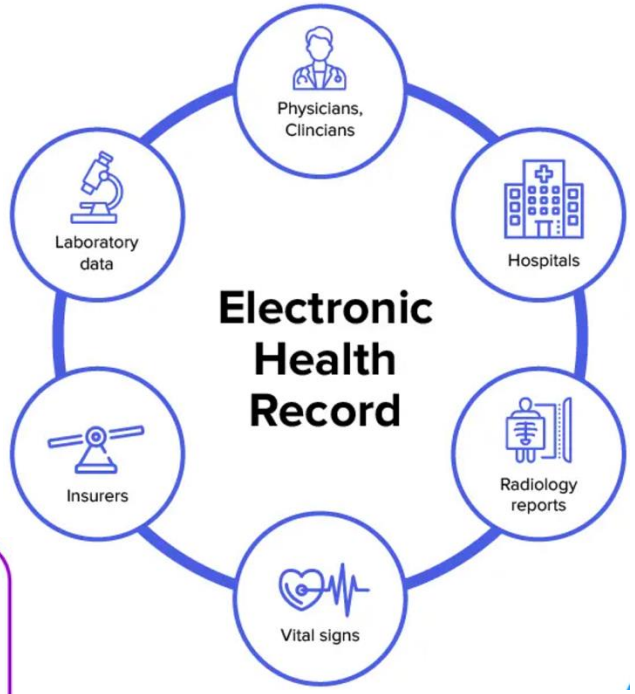


“Datafication” of life sciences

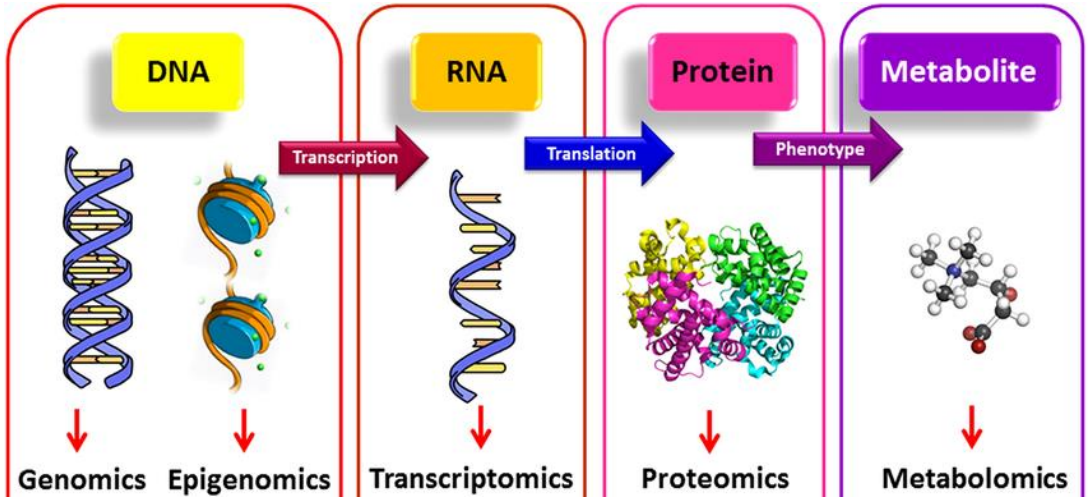
Imaging



Clinical data

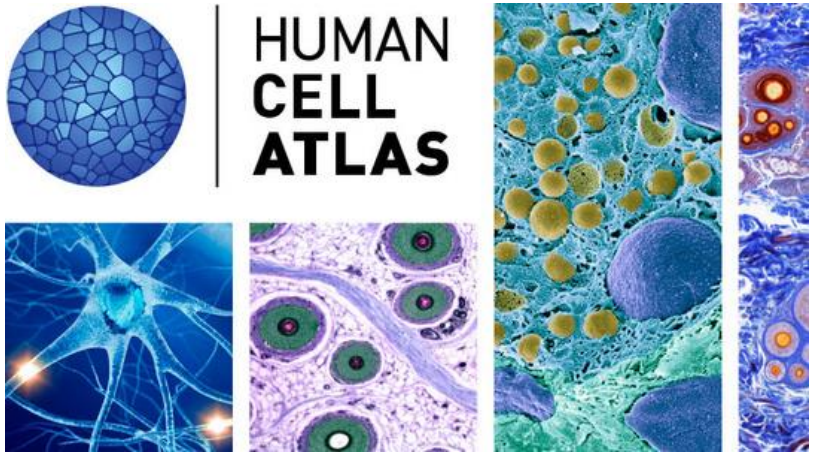


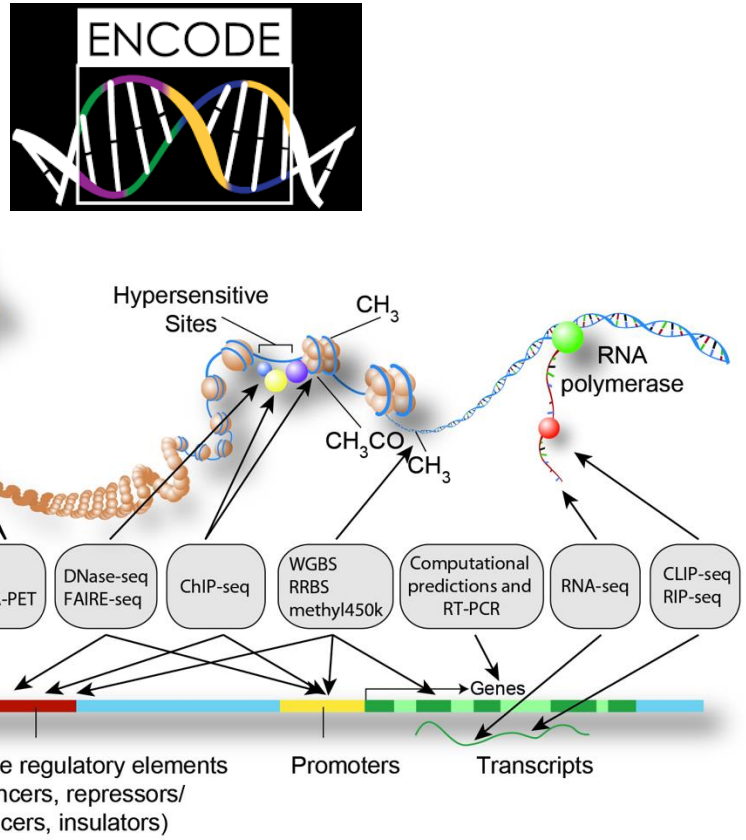
Omics

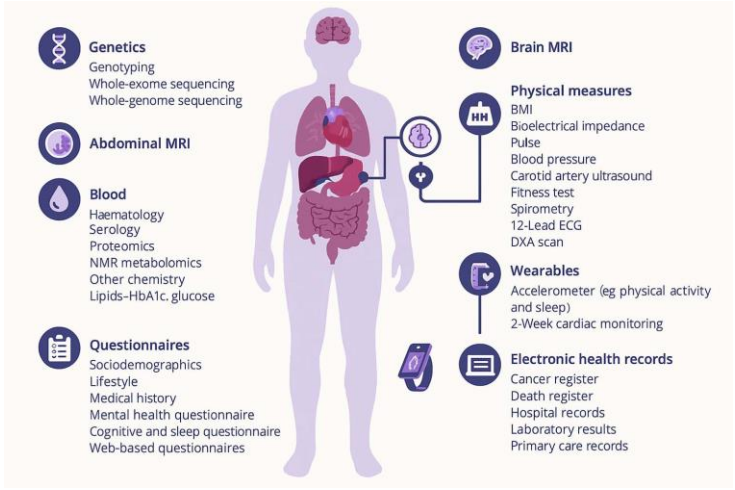


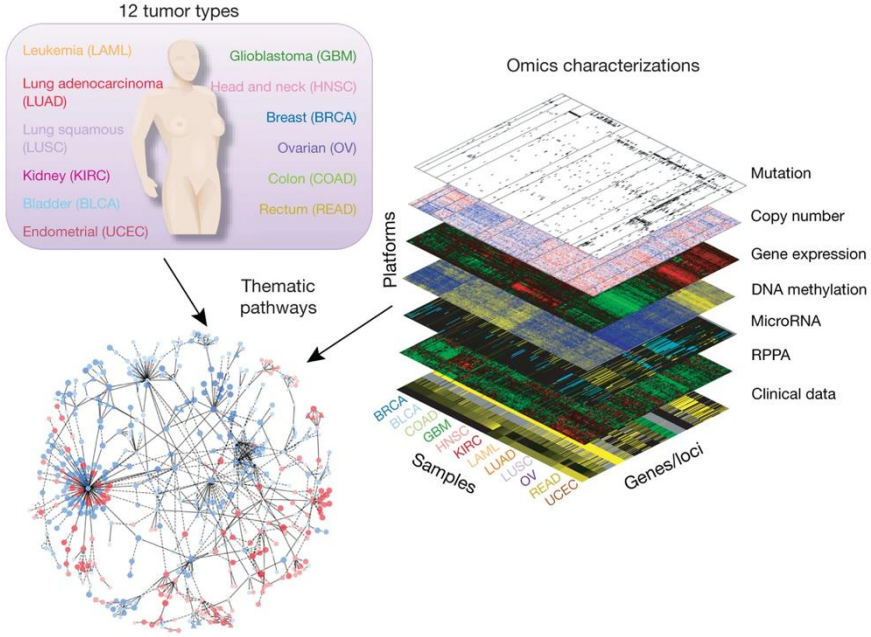
Wearable health data

Big omics datasets

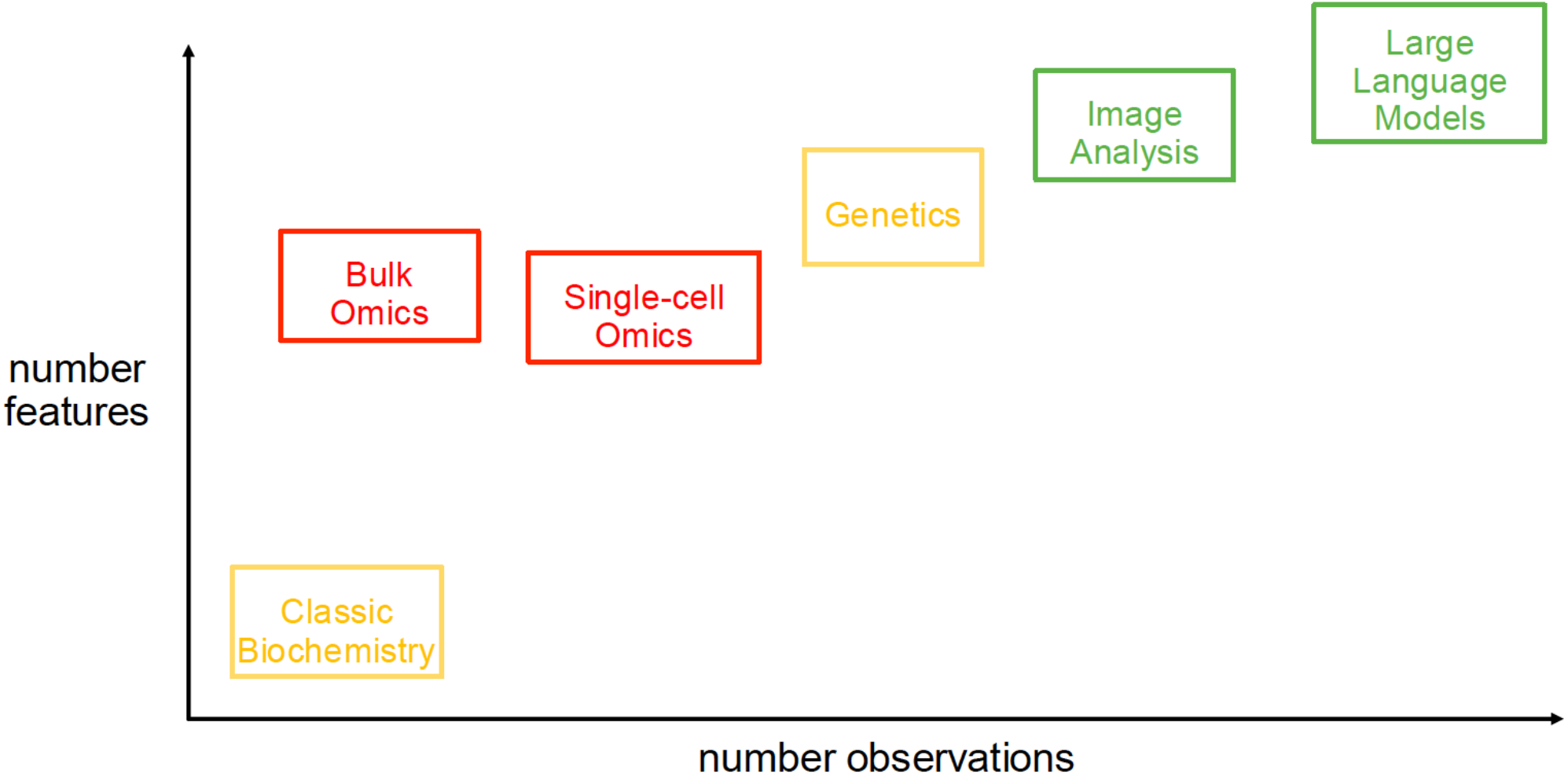








Biomedical data is high-dimensional



Slide courtesy of Martin Garrido-Rodriguez

What is high dimensional data

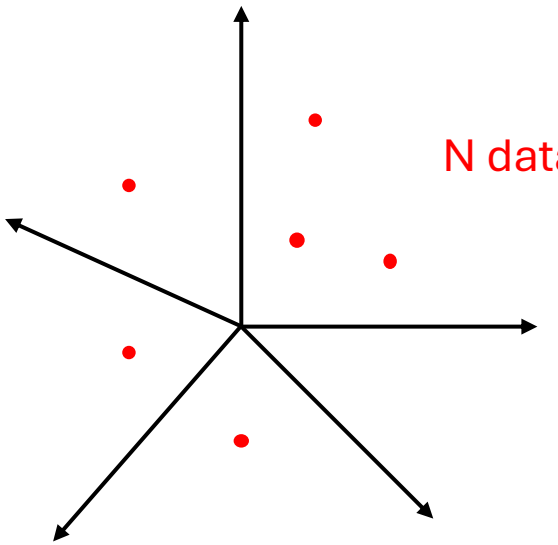
Features: genes, proteins, microbes, metabolites etc. ← P

N

Statistical observations: e.g. samples, cells etc.

$$\begin{bmatrix} 0 & 8 & 34 & 85 & 44 \\ 1 & 4 & 6 & 35 & 31 \\ 8 & 6 & 78 & 8 & 92 \\ 10 & 3 & 23 & 5 & 3 \\ 6 & 45 & 98 & 76 & 8 \\ 32 & 8 & 23 & 84 & 0 \end{bmatrix}$$

P-dimensional space



N data points

For a robust statistical analysis, one should properly "sample" the P-dimensional space, hence large sample size is required, **N >> P**

In biology, N << P

- Note: samples N (should be independent)

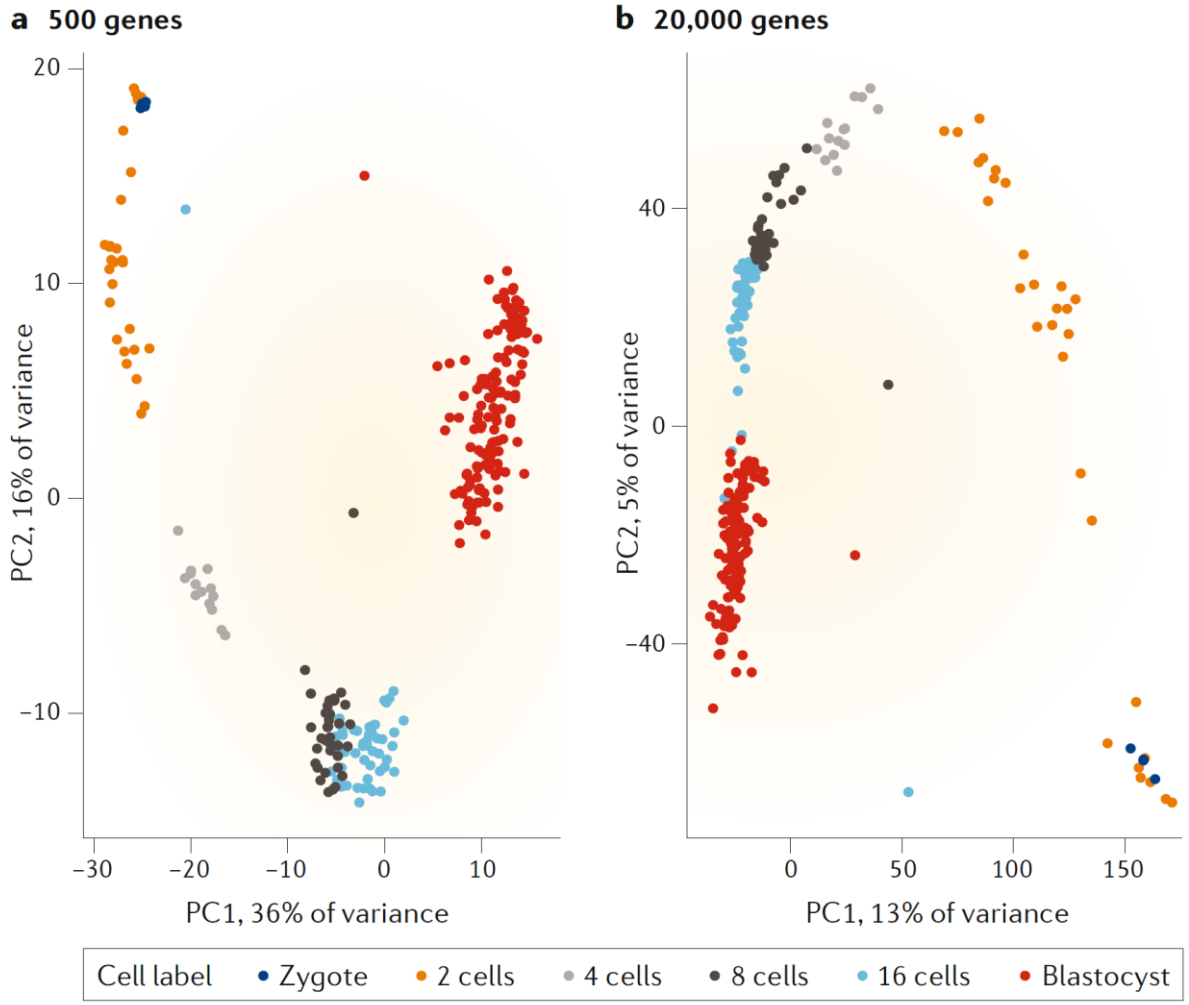
- We often have less samples than features
  - Example: human genome has ~ 20k genes, but very few studies have gene expression measurements in 20k samples

- When N << P → **Curse of dimensionality**

Curse of dimensionality

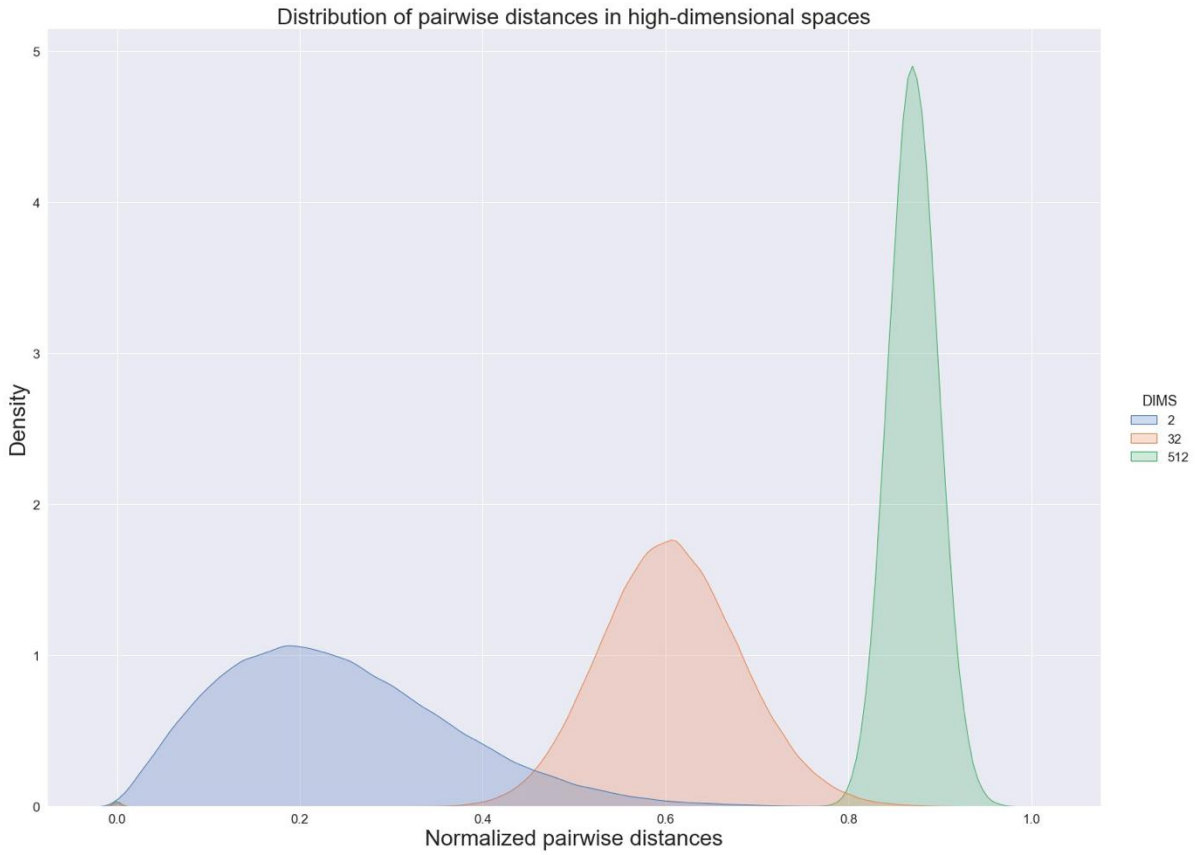
- More features (genes) -> less information (worse separation of classes)!!!

- Remember, biological measurements are VERY noisy

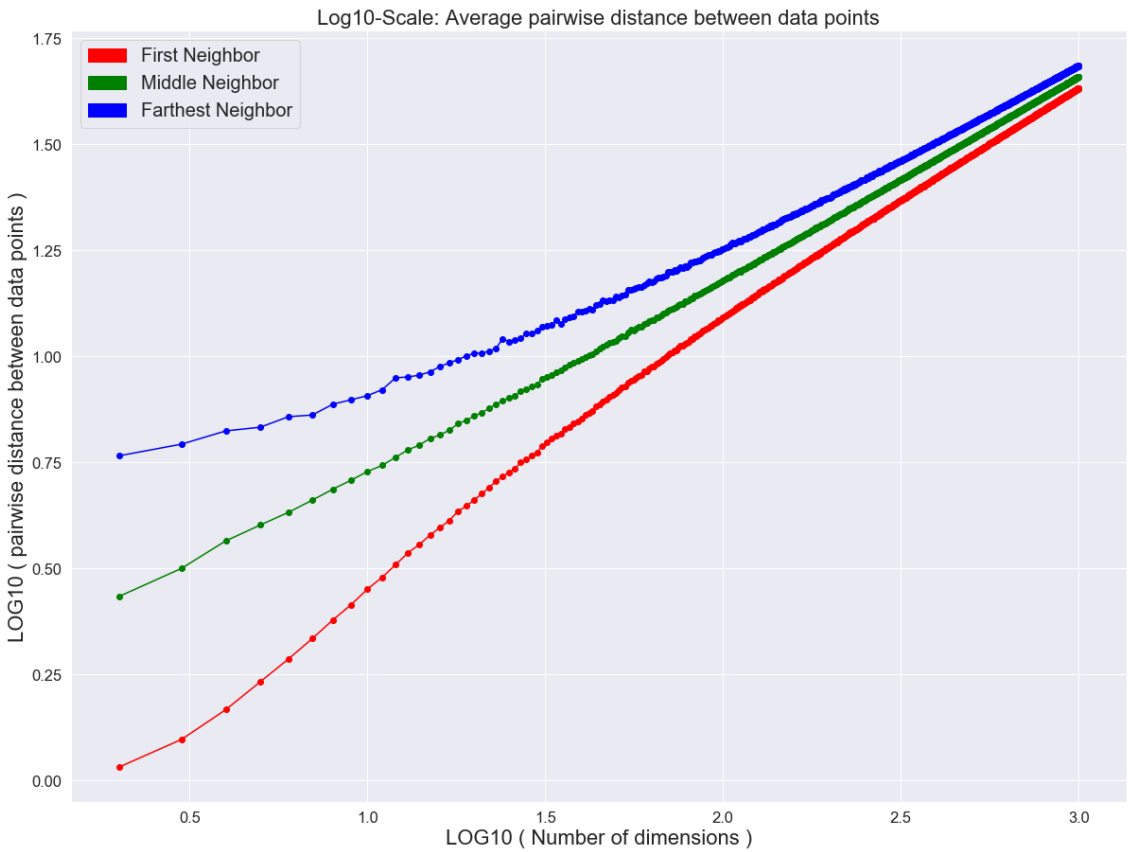


Curse of dimensionality

Distances in high-dimensional space

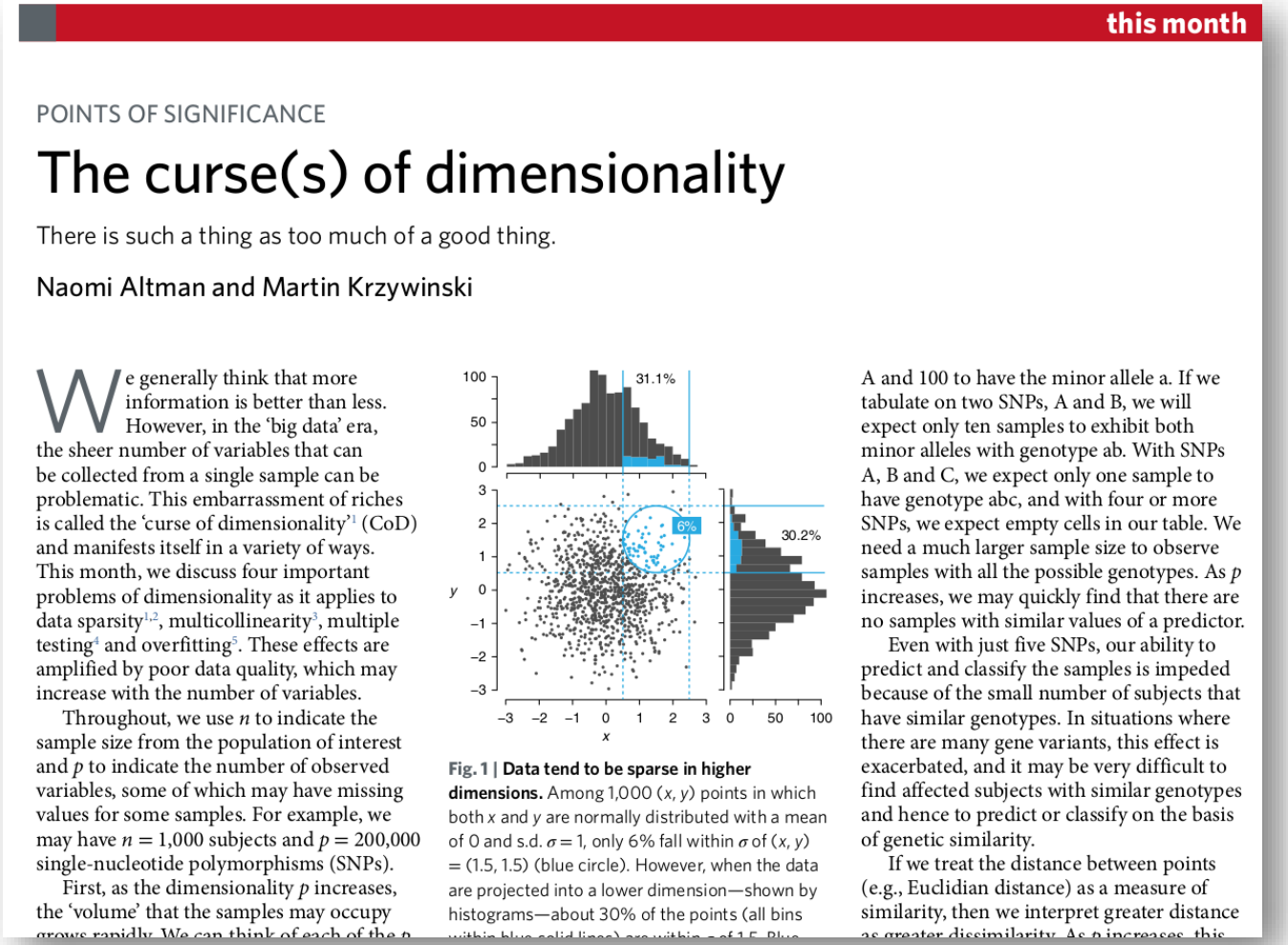


Data points become far from each other and equidistant in high dimensions



The differences between closest and farthest data point neighbours disappears in high-dimensional spaces: can't run cluster analysis

Curse of dimensionality



this month

POINTS OF SIGNIFICANCE

The curse(s) of dimensionality

There is such a thing as too much of a good thing.

Naomi Altman and Martin Krzywinski

We generally think that more information is better than less. However, in the 'big data' era, the sheer number of variables that can be collected from a single sample can be problematic. This embarrassment of riches is called the 'curse of dimensionality'[1] (CoD) and manifests itself in a variety of ways. This month, we discuss four important problems of dimensionality as it applies to data sparsity[1,2], multicollinearity[3], multiple testing[4] and overfitting[5]. These effects are amplified by poor data quality, which may increase with the number of variables.

Throughout, we use n to indicate the sample size from the population of interest and p to indicate the number of observed variables, some of which may have missing values for some samples. For example, we may have $n = 1{,}000$ subjects and $p = 200{,}000$ single-nucleotide polymorphisms (SNPs).

First, as the dimensionality p increases, the 'volume' that the samples may occupy grows rapidly. We can think of each of the p

Fig. 1 | Data tend to be sparse in higher dimensions. Among 1,000 (x, y) points in which both x and y are normally distributed with a mean of 0 and s.d. $\sigma = 1$, only 6% fall within σ of $(x, y) = (1.5, 1.5)$ (blue circle). However, when the data are projected into a lower dimension—shown by histograms—about 30% of the points (all bins within blue solid lines) are within σ of 1.5. Blue

A and 100 to have the minor allele a. If we tabulate on two SNPs, A and B, we will expect only ten samples to exhibit both minor alleles with genotype ab. With SNPs A, B and C, we expect only one sample to have genotype abc, and with four or more SNPs, we expect empty cells in our table. We need a much larger sample size to observe samples with all the possible genotypes. As p increases, we may quickly find that there are no samples with similar values of a predictor.

Even with just five SNPs, our ability to predict and classify the samples is impeded because of the small number of subjects that have similar genotypes. In situations where there are many gene variants, this effect is exacerbated, and it may be very difficult to find affected subjects with similar genotypes and hence to predict or classify on the basis of genetic similarity.

If we treat the distance between points (e.g., Euclidian distance) as a measure of similarity, then we interpret greater distance as greater dissimilarity. As p increases, this

Altman N, Krzywinski M. The curse(s) of dimensionality. Nat Methods. 2018 Jun;15(6):399-400. doi: 10.1038/s41592-018-0019-x. PMID: 29855577.

Dimensions of biomedical omics data

$$\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix}$$

Proteomics
Metabolomics
Metagenomics

$N \approx P$

$$\begin{pmatrix} 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 \end{pmatrix}$$

Transcriptomics

$N << P$

$$\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 \end{pmatrix}$$

Genomics (GWAS, WGS, methylation, etc.)

$N <<<< P$

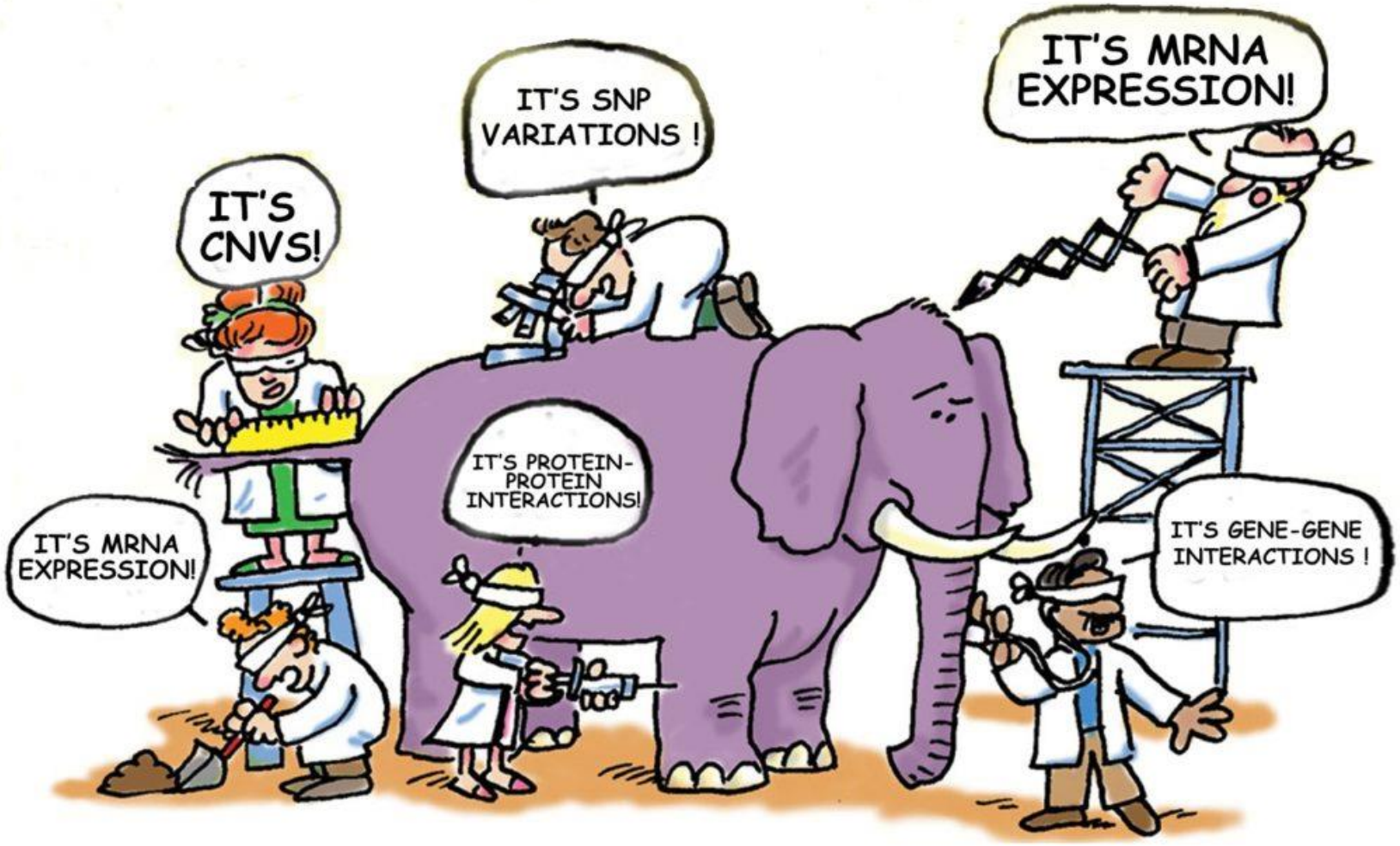
High-dimensional data analysis

- **Dimensionality reduction** (representation learning) → visualization, reduce noise, computational efficiency

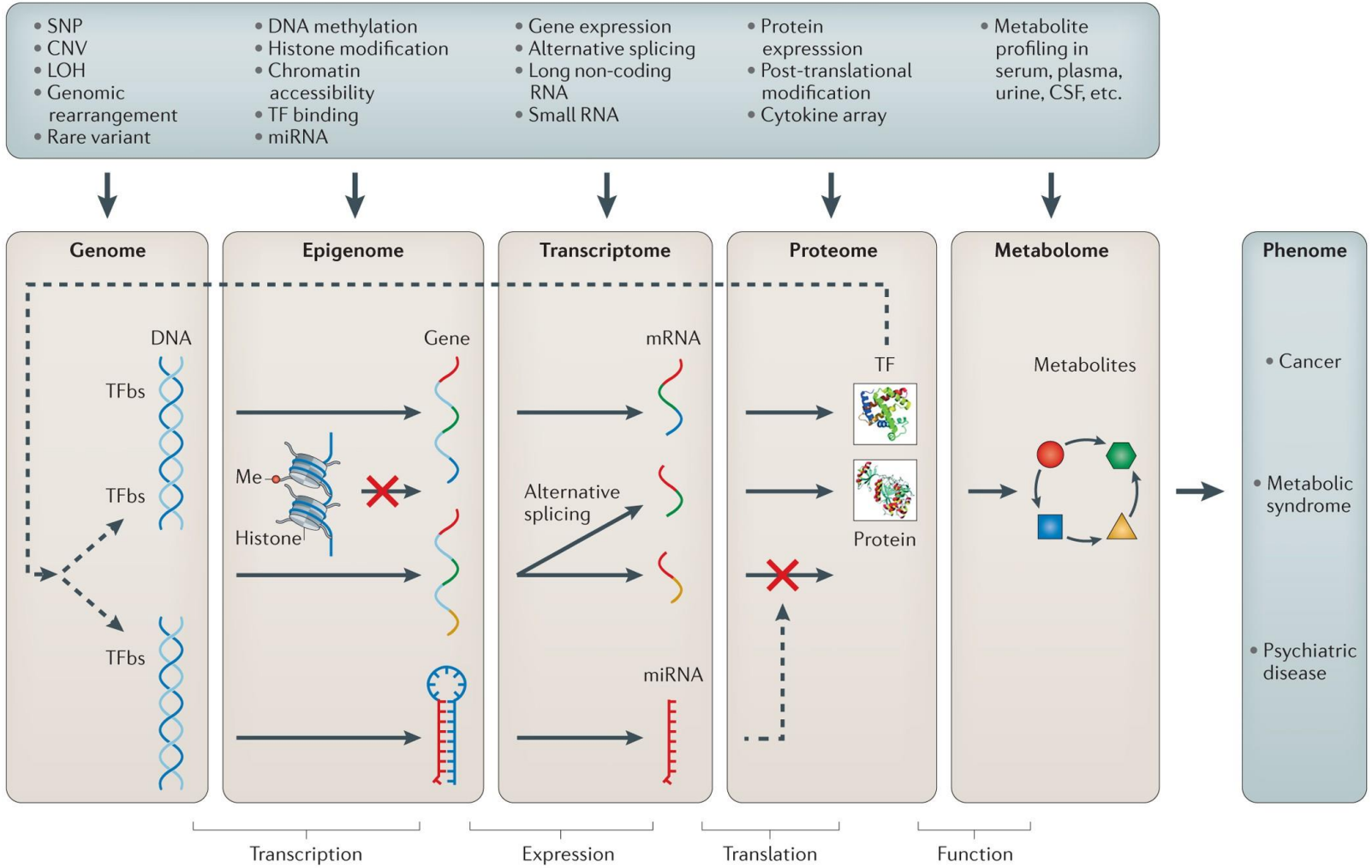
- **Clustering** (unsupervised learning) → identifying groups of similar samples

- **Classification** (supervised learning) → predicting outcomes, identifying informative features

Biological data is increasingly multi-omics



Multi-omics data provides complementary information



Ritchie et al, Nature review Genetics 2015

Nature Reviews | Genetics

Multi-omics data integration, the **WHAT** ?

Data integration is the process of combining different types of omic data as predictor variables to allow comprehensive modelling of complex traits or phenotypes

Ritchie et al, Nature review Genetics 2015

Multi-omics data integration, the **WHY** ?

- Biology is complex and processes happen across molecular layers. Integration gives a holistic view of biological processes

- **Fundamental:** understanding interactions across molecular layers (e.g. how is protein expression regulated?)

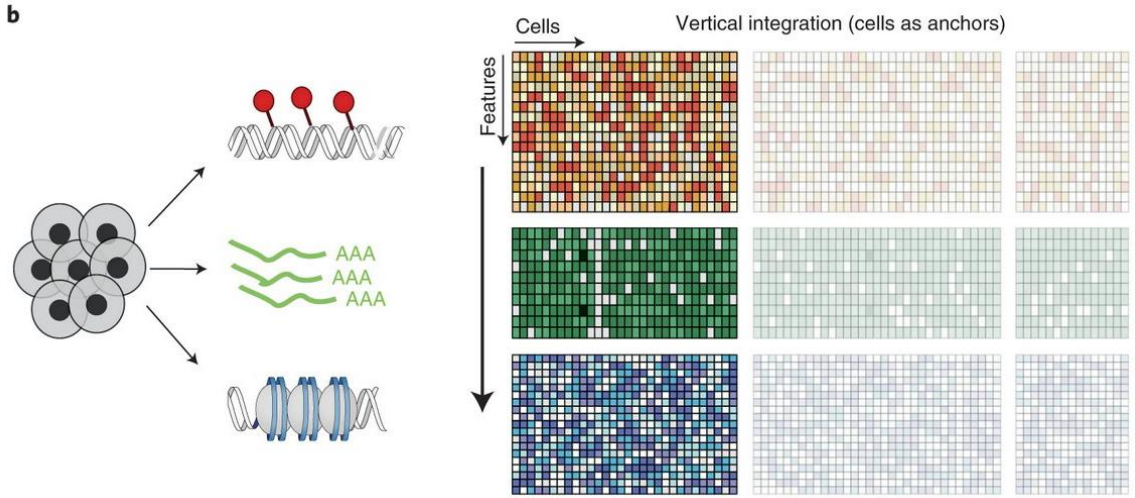
- **Translational/clinical:** what are the best biomarkers of disease progression/patient stratification/…?

Multi-omics data integration, the **HOW**?

Horizontal integration
(across samples)

Same data modality profiled from
independent groups of samples

Vertical integration
(across features)



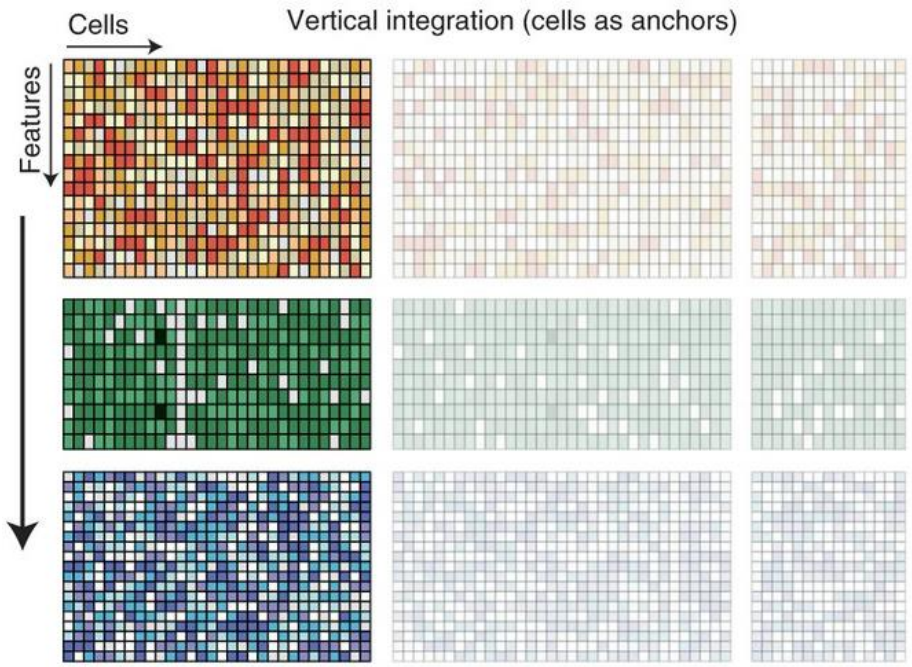
Multiple data modalities profiled
from the same sample

Argelaguet et al., Nature Biotechnology 2021

Vertical data integration approaches

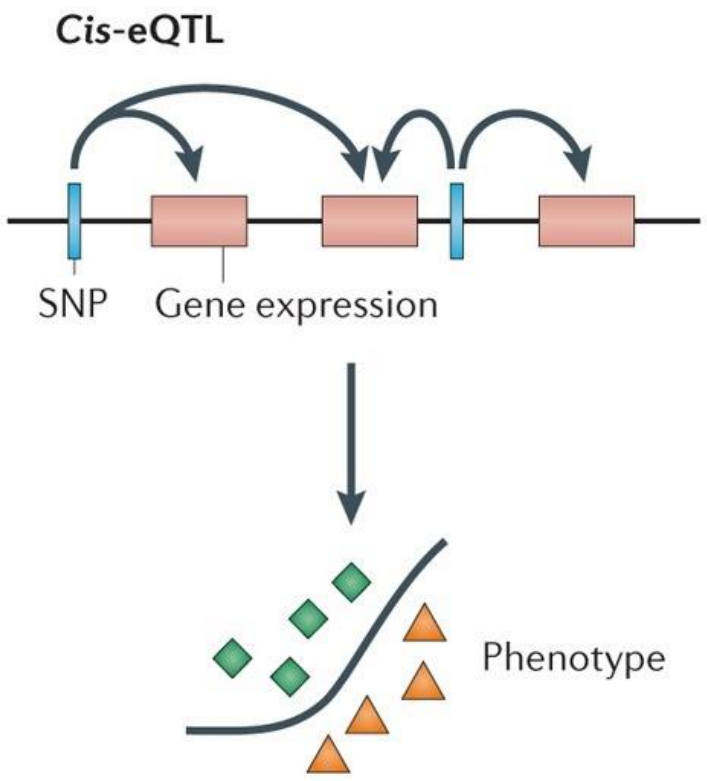
- Feature-based analysis:
  - Association analysis between features from different data modalities
  - Gene regulatory networks

- Sample/Cell-based analysis (global integration)



Argelaguet et al., Nature Biotechnology 2021

I. Association across features



phenotype

- GWAS
trait or disease
- eQTL
bulk gene expression

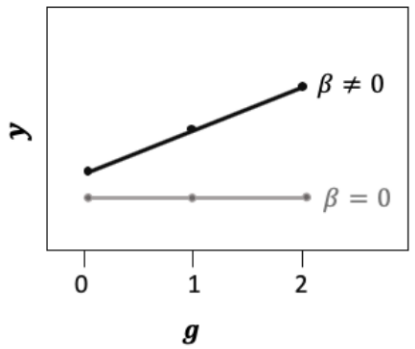
SNP effect

$$\underset{N}{\boldsymbol{y}} = \underset{N}{\boldsymbol{g}}\beta + \underset{N}{\boldsymbol{\varepsilon}}$$

SNP

noise

$\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma_n{}^2 \boldsymbol{I})$



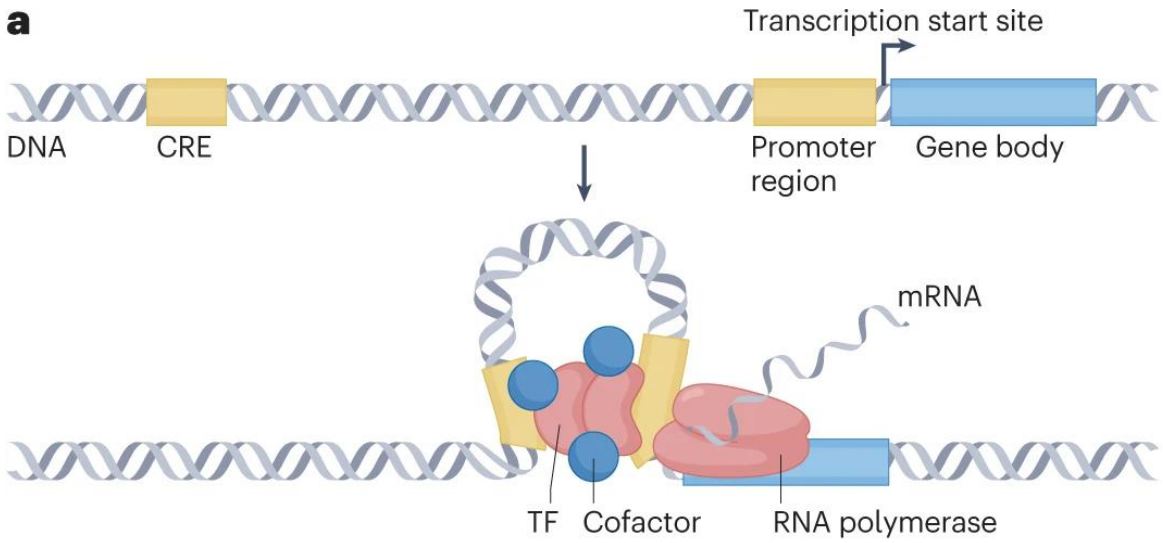
Test:
H0: $\beta = 0$
H1: $\beta \neq 0$

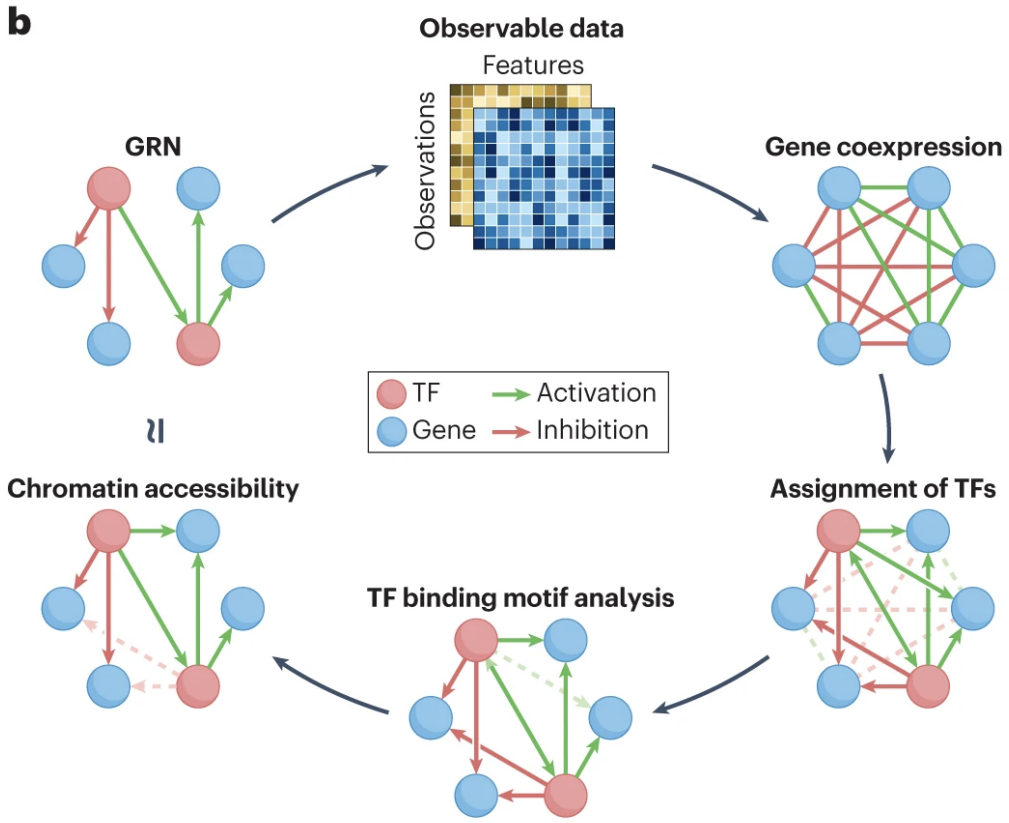
GWAS: Genome-Wide Association Study (GWAS)
eQTL: expression Quantiative Trait Loci

Figure courtesy of Ricardo Ramirez and Ricard Argelaguet

Ritchie et al, Nature review Genetics 2015

II. Gene regulatory networks





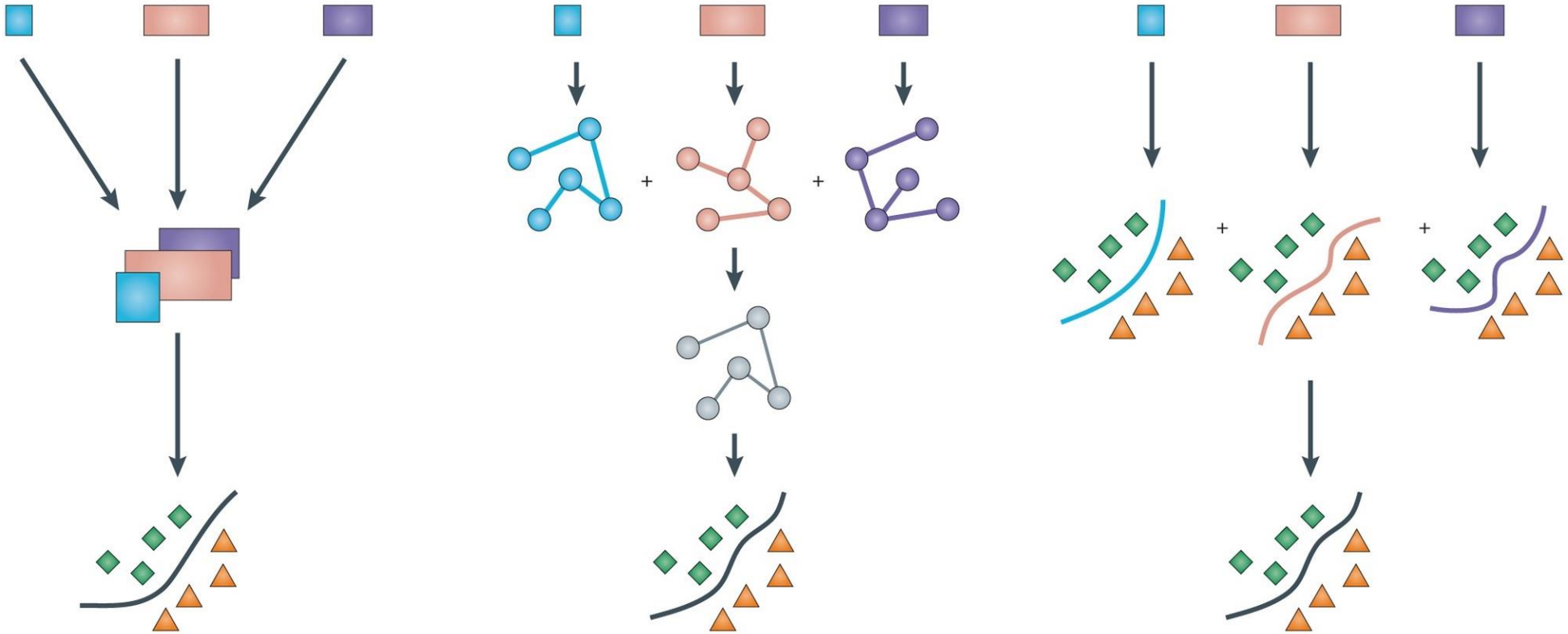
Badia-i-Mompel, et. al., JAHA, 2021

III. Global vertical integration

Early integration
(concatenation-based)

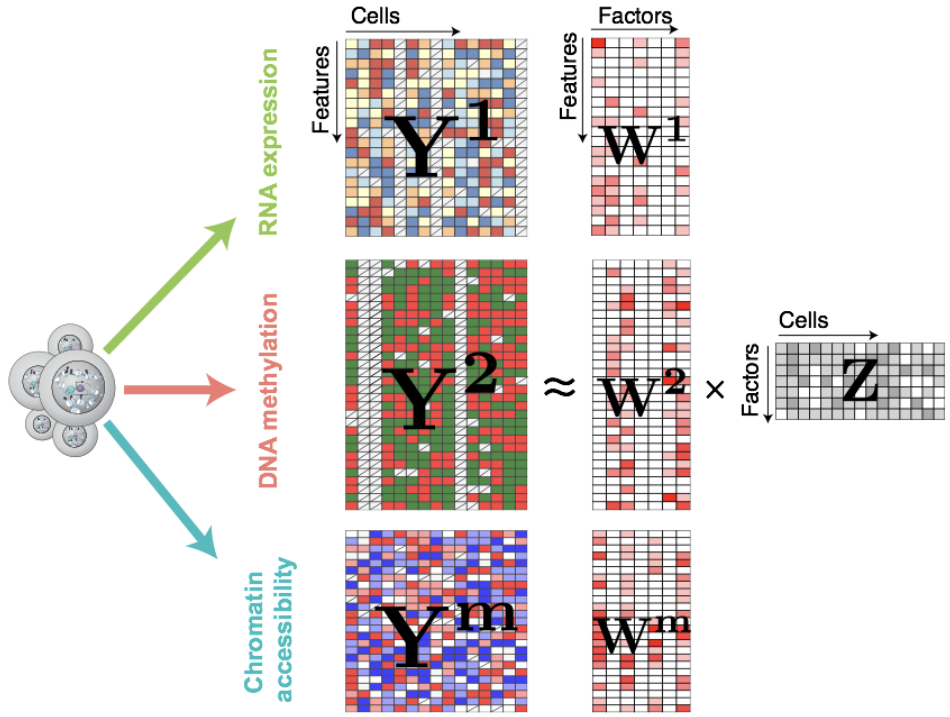
Mid integration
(transformation-based)

Late integration
(model-based)

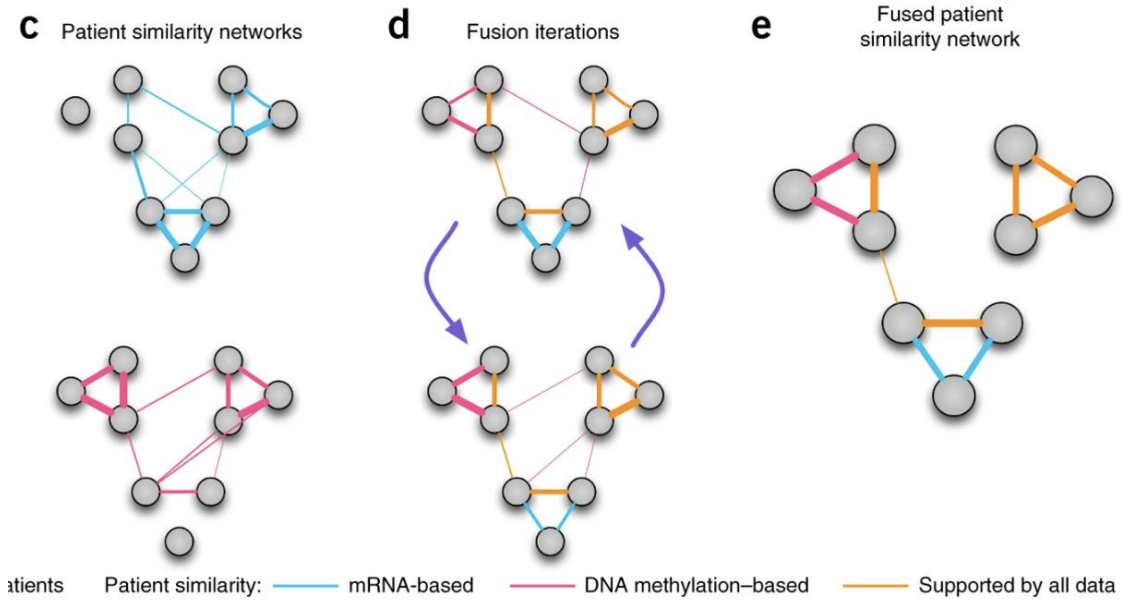


Ritchie et al, Nature review Genetics 2015

Examples of vertical data integration



Multi-Omics Factor Analysis (MOFA)
Argelaguet & Velten et al. *Mol Sys Bio* (2018)



Similarity Network Fusion
Wang et al. *Nature Methods* (2014)

Challenges of vertical integration

- Data collected using different techniques (i.e. data modalities) generally exhibit heterogeneous statistical properties
- Large amounts (and different patterns) of missing values
- Overfitting
- Undesired sources of heterogeneity
- Complexity of the data requires unsupervised interpretable approaches

Types of integration methods

	Linear	**Non-linear**
Unsupervised	PCA, Factor analysis (NMF, MOFA,…),…	Autoencoders, kernel PCA,…
Supervised	LASSO, Ridge, Elastic-Net, PLS-DA, OPLS,…	Bayesian networks, classifiers (e.g. Random Forest), multi-layer perceptron,…

- Do I have a phenotype of interest, or do I want to explore my data?
- Do I expect non-linear relationships between features?
- Do I have enough data?

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Resources

- **Methods of integrating data to uncover genotype–phenotype interactions.** Ritchie et al, Nature review Genetics 2015

- **A technical review of multi-omics data integration methods: from classical statistical to deep generative approaches.** Baião et al. Briefings in Bioinformatics 2025

- Computational principles and challenges in single-cell data integration. Argelaguet et al. Nature Biotechnology 2021

Day 1 (Wednesday 22.04.2026)

Time	Title
09:00 - 09:15	Workshop introduction + housekeeping
09:15 - 10:00	(1) Introduction to data integration
10:00 - 10:15	*Break*
10:15 - 11:15	(2) Case study data overview + CyTOF data processing tutorial
11:15 - 12:00	(3) Multi-omics research at UVRI
12:00 - 13:00	*Lunch*
13:00 - 14:00	(4) Lecture — PCA + PLS-DA
14:00 - 14:45	(5) Practical — PCA
14:45 - 15:00	*Break*
15:00 - 15:45	(6) Practical — PLS-DA
15:45 - 16:00	Workshop recap and day 2 preview

Day 2 (Thursday 23.04.2026)

Time	Title
09:00 - 10:00	(1) Introduction to Single-cell Genomics
10:00 - 10:15	*Break*
10:15 - 11:00	(2) Lecture – DIABLO
11:00 - 12:00	(3) Practical – DIABLO
12:00 - 13:00	*Lunch*
13:00 - 13:45	(4) Lecture — MOFA
13:45 - 14:45	(5) Practical — MOFA
14:45 - 15:00	*Break*
15:00 - 15:30	(6) Practical — MOFA (continued)
15:30 - 16:00	Workshop closing + Q&A

Course materials:

https://github.com/mahfouzlab/UVRI-workshop-2026

Practical:

Posit (you received an invite, don’t share it)